

# An enthalpy–entropy competition strategy enables moisture-stable and scalable perovskite photovoltaics

Received: 21 December 2025

Accepted: 21 July 2026

Published online: 17 August 2026

 Check for updates

Zhaoyang Chu<sup>1,6</sup>, Junliang Li<sup>1,6</sup>, Jiaju Gao<sup>1,6</sup>, Sihao Li<sup>1</sup>, Baojin Fan<sup>2</sup>, Zongyi Gu<sup>1</sup>, Yihuan Xie<sup>1</sup>, Yaling Luo<sup>1</sup>, Zhi Xing<sup>3</sup>, Xiangchuan Meng<sup>1</sup>, Hongxiang Li<sup>4</sup>, Xiaotian Hu<sup>1,5</sup>✉ & Yiwang Chen<sup>1,2,3,5</sup>✉

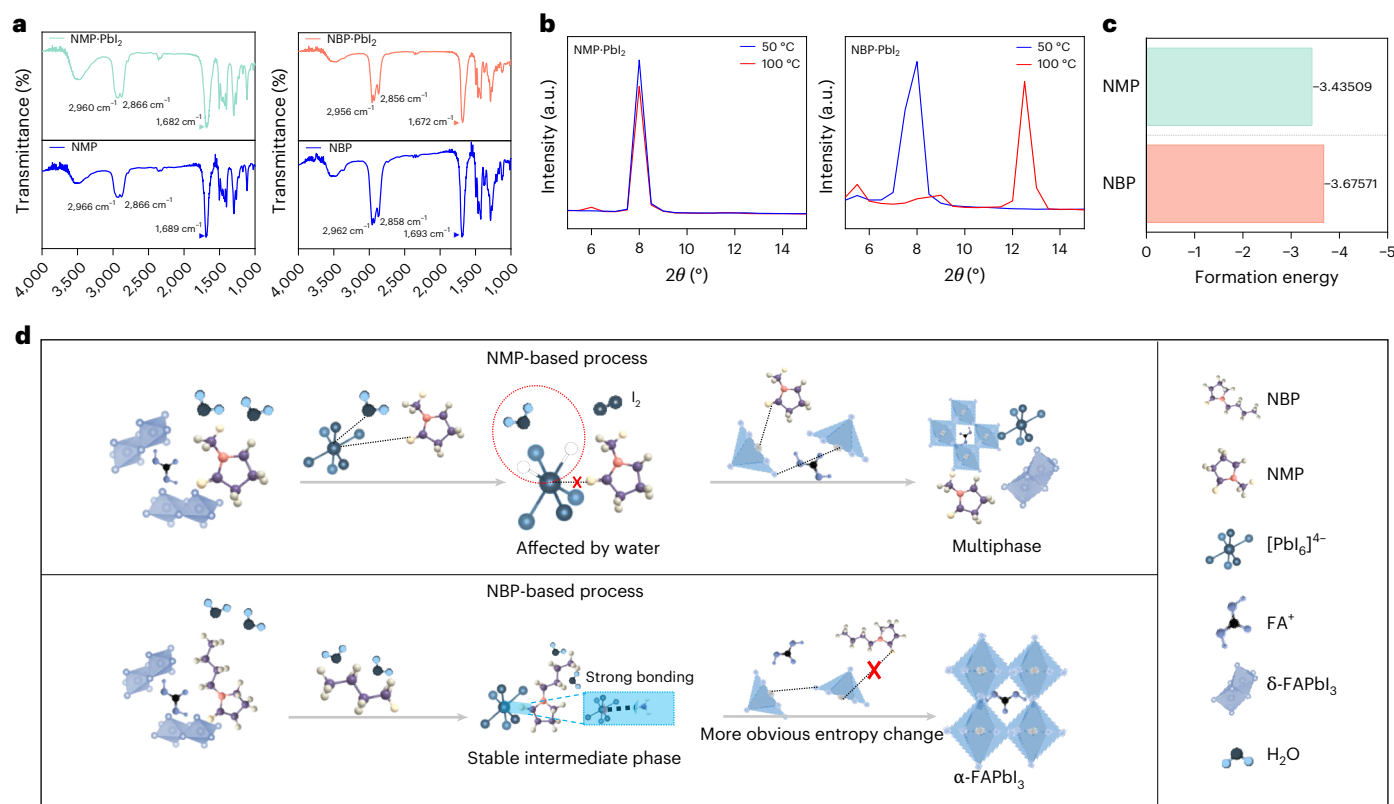
Scalable fabrication of perovskite solar modules is hindered by processing inconsistencies and moisture-induced degradation. Although classic coordination solvents form thermodynamically stable intermediate phases with lead iodide (PbI<sub>2</sub>) under inert gas atmospheres, they are highly hygroscopic in ambient conditions. Moreover, due to minimal entropy change, this phase is difficult to remove during annealing, leading to micropores and unwanted phases. Here we introduce an enthalpy–entropy competition strategy for scalable, humidity-tolerant perovskite printing, in which dimethyl sulfoxide/*N*-methylpyrrolidone is replaced by *N*-butylpyrrolidone. *N*-butylpyrrolidone forms strong carbonyl–PbI<sub>2</sub> coordination at ambient temperatures (enthalpy-driven), yet readily dissociates on heating (entropy-driven), facilitating easier formamidinium iodide embedding and intermediate-phase protection. The strategy yields perovskite solar modules with certified power conversion efficiencies of 23.97% in a 100-cm<sup>2</sup> rigid module and 19.71% in a flexible counterpart. Encapsulated devices retain 80% of initial performance after 1,440 hours under the double 85 condition (85 °C, 85% relative humidity), demonstrating outstanding operational durability.

The certified power conversion efficiency (PCE) of perovskite solar cells (PSCs) has exceeded 27%, meeting commercialization benchmarks<sup>1,2</sup>. However, such high-performance devices are predominantly fabricated via spin coating under inert atmospheres<sup>3–5</sup>. During the transition from laboratory-scale to commercial-scale fabrication, solution printing processes represent a critical pathway towards achieving manufacturing scalability<sup>6–9</sup>. So far, binary solvent systems combining volatile solvents (for example, dimethylformamide, 2-methoxyethanol) with high-boiling-point coordination solvents (such as dimethyl sulfoxide

(DMSO), *N*-methylpyrrolidone (NMP)) are widely adopted<sup>10–14</sup>. Through Lewis acid–base coordination interactions, DMSO and NMP form intermediates such as DMSO–PbI<sub>2</sub> and NMP–PbI<sub>2</sub>–formamidinium iodide (FAI) with lead iodide (PbI<sub>2</sub>)<sup>15–17</sup>. During the crystallization process, FAI is embedded in situ to promote ordered crystallization, thereby forming dense perovskite films and reducing the formation of undesired phases (such as  $\delta$ -FAPbI<sub>3</sub>)<sup>18–20</sup>.

As polycrystalline materials, the crystallographic quality of perovskites, such as crystal orientation and facet ordering, plays a pivotal

<sup>1</sup>College of Chemistry and Chemical Engineering, Film Energy Chemistry for Jiangxi Provincial Key Laboratory, Institute of Polymers and Energy Chemistry, Nanchang University, Nanchang, China. <sup>2</sup>College of Chemistry and Chemical Engineering, Key Laboratory of Fluorine and Silicon for Energy Materials and Chemistry of Ministry of Education, Jiangxi Normal University, Nanchang, China. <sup>3</sup>College of Chemistry and Materials, Gannan Normal University, Ganzhou, China. <sup>4</sup>College of Materials and Energy, Guang'an Institute of Technology, Guang'an, China. <sup>5</sup>Key Laboratory for Advanced Optoelectronic Integrated Chips of Jiangsu Province, Peking University Yangtze Delta Institute of Optoelectronics, Nantong, China. <sup>6</sup>These authors contributed equally: Zhaoyang Chu, Junliang Li, Jiaju Gao. ✉ e-mail: [happyhu@ncu.edu.cn](mailto:happyhu@ncu.edu.cn); [ywchen@ncu.edu.cn](mailto:ywchen@ncu.edu.cn)



**Fig. 1 | Determination of enthalpy–entropy effects.** **a**, Fourier-transform infrared spectroscopy of NMP, NMP-lead iodide (PbI<sub>2</sub>), NBP and NBP-PbI<sub>2</sub>. **b**, XRD peak changes in NMP-PbI<sub>2</sub> and NBP-PbI<sub>2</sub> at 50 °C and 100 °C. **c**, Perovskite

formation energy formed by NMP and NBP and calculated by the density functional theory. **d**, Schematic diagram of the thermodynamic driving force and kinetics of the NBP enthalpy–entropy competition strategy.

role in determining both device efficiency and operational stability<sup>21–23</sup>. However, during the scale-up from small-area devices to large-area perovskite solar module (PSM) fabrication, perovskite nucleation and crystallization processes face dual challenges from process transitions and exposure to moisture (H<sub>2</sub>O)<sup>24–26</sup>. High-boiling-point solvents commonly used in scalable printing exhibit strong hygroscopicity, which holds back solvent evaporation and promotes the formation of hydrated intermediates<sup>27–30</sup>. This leads to incomplete conversion of intermediate complexes into perovskite, generating numerous pinholes and defects. Concurrently, the high energy barrier for α-FAPbI<sub>3</sub> phase formation and the complexity of printing-induced crystallization can cause mixed phases (2H, 4H, 6H and so on) in perovskite films<sup>31,32</sup>. Moreover, the (100) crystal plane and surface vacancies of FAPbI<sub>3</sub> crystals are also water adsorption sites, which further aggravates the perovskite degradation and accelerates the δ-FAPbI<sub>3</sub> phase formation<sup>33,34</sup>. Thus, it remains critical to develop strategies that simultaneously protect intermediate phases, lower the energy barrier for α-phase formation and precisely regulate nucleation and growth pathways under ambient conditions to yield compact, oriented and phase-pure perovskite films.

Here we introduce an enthalpy–entropy competition strategy for scalable, humidity-tolerant perovskite printing, in which DMSO/NMP is replaced by the low-hygroscopic solvent *N*-butylpyrrolidone (NBP). The dynamic behaviour of its butyl group enables the carbonyl group to form dynamic coordination bonds with PbI<sub>2</sub>, characterized by a more negative enthalpy change (Δ*H*) but accompanied by obvious entropy (Δ*S*) loss. At room temperature, the enthalpy effect dominates: the bulky butyl group induces a more dispersed electron cloud distribution in NBP, increasing the molecular dipole moment. This strengthens electrostatic interactions between NBP and PbI<sub>2</sub>, facilitating the formation of more stable coordination bonds, prevent the intermediate phase from being compromised by water. At elevated temperatures,

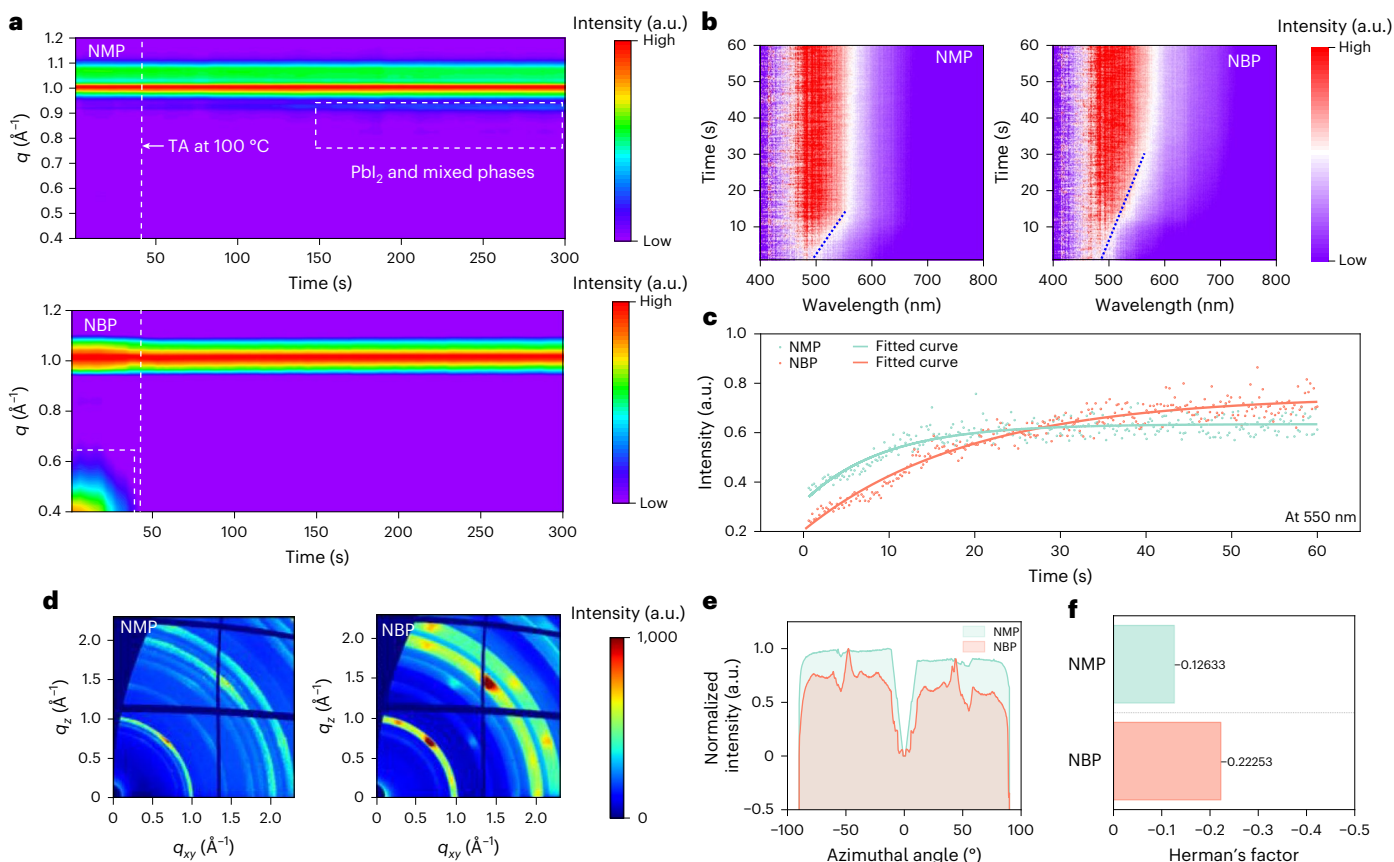
the thermal motion of molecules is intensified. The large butyl group in NBP, due to its substantial steric hindrance, is more prone to collisions and disturbances from other molecules during thermal motion. This enhances the contribution of the entropy term (–*T*Δ*S*), accelerating the dissociation reaction. As a result, FAI can be more readily embedded into PbI<sub>2</sub>, reducing the energy barrier for α-FAPbI<sub>3</sub> phase formation. Small-area devices fabricated with NBP achieve a PCE of 26.50%, as the aperture area increases (from 4 cm<sup>2</sup> to 100 cm<sup>2</sup>), the PCE loss becomes smaller, only 2.61% is lost at 100 cm<sup>2</sup>. In addition, rigid PSMs demonstrate a PCE of 24.15%, while flexible PSMs achieve a PCE of 20.6%.

## Results and discussion

### Determination of enthalpy–entropy effects

Owing to its strong hygroscopicity (Supplementary Fig. 1), DMSO is unsuitable for scalable, moisture-tolerant printing and is typically substituted with NMP. Furthermore, we compared the humidity stability of two perovskite inks stored at 60% relative humidity for 24 hours and found that NBP exhibited superior moisture resistance (Supplementary Fig. 2). The chemical structures of NMP and NBP are shown in Supplementary Fig. 3a. Density functional theory calculations reveal that, compared with NMP, the carbonyl group of NBP exhibits higher electronegativity, attributed to the electron-withdrawing effect of the butyl group, which redistributes the carbonyl electron density and enhances the molecular dipole moment (Supplementary Fig. 3b). Experimentally, compared with NBP, when using NMP for dissolution, there is still some residual PbI<sub>2</sub> (Supplementary Fig. 4).

Fourier-transform infrared spectroscopy (Fig. 1a) further confirms stronger coordination between NBP and PbI<sub>2</sub>. The C=O stretching vibration of NMP shifts modestly from 1,689 cm<sup>-1</sup> to 1,682 cm<sup>-1</sup> on complexation, while the C–H stretching band narrows from 2,866–2,966 cm<sup>-1</sup> to 2,866–2,960 cm<sup>-1</sup>. By contrast, NBP exhibits a larger C=O redshift,



**Fig. 2 | Nucleation and crystallization regulation.** **a**, Time evolution of GIWAXS intensity along the  $q_z$  direction for the NMP sample and NBP sample. **b**, Time-resolved in situ UV-visible (UV-vis) absorption spectra during thermal annealing process. **c**, UV-vis absorption intensity of different perovskite films extracted at  $\approx 550$  nm wavelength as a function of thermal annealing time. **d**, GIWAXS patterns

from  $1,693 \text{ cm}^{-1}$  to  $1,672 \text{ cm}^{-1}$ , suggesting enhanced Pb–O coordination. In addition, only minor narrowing of the C–H band is observed ( $2,962\text{--}2,866 \text{ cm}^{-1}$  to  $2,956\text{--}2,856 \text{ cm}^{-1}$ ), supporting the presence of both coordination and hydrogen bonding. Temperature-dependent X-ray diffraction (XRD) analysis (Fig. 1b) shows that both NMP and NBP form stable complexes with PbI<sub>2</sub> at 50 °C (characterized by a distinct peak at 8 °C). However, on heating to 100 °C, only NBP exhibits dissociation from PbI<sub>2</sub>, indicating thermally responsive ligand dynamics. This temperature-driven release can be rationalized using the Gibbs free energy equation:

$$\Delta G = \Delta H - T\Delta S$$

where  $\Delta G$  represents the change in Gibbs free energy,  $\Delta H$  is the enthalpy change,  $T$  is the temperature and  $\Delta S$  is the entropy change. To elucidate the thermodynamic contributions governing NBP–PbI<sub>2</sub> interactions, we performed density functional theory calculations. The results indicate that NBP exhibits a more negative adsorption energy with PbI<sub>2</sub> compared with NMP (Supplementary Fig. 5), consistent with stronger coordination and a dominant  $\Delta H$  contribution at ambient conditions. However, during thermal annealing, NBP leads to a lower formation energy for the perovskite phase (Fig. 1c), suggesting that entropy gain ( $\Delta S$ ) increasingly drives the process at elevated temperatures.

On the basis of these observations, we propose a temperature-dependent coordination mechanism for NBP (Fig. 1d). At room temperature, the carbonyl group of NBP forms strong coordination bonds with PbI<sub>2</sub> ( $\Delta H$ -dominated), stabilizing the intermediate phase and impeding moisture-induced degradation. As temperature increases, the

of the NMP and NBP perovskite films. **e**, Integrated scattering intensity plots of the NMP and NBP perovskite films azimuthally along the ring at the total in-plane wave vector,  $q_z = 1.0 \text{ \AA}^{-1}$ , assigned to the (100) facet. **f**, Comparison of Herman's orientation factor of the NMP and NBP perovskite films.

sterically hindered butyl group facilitates dynamic bond dissociation ( $\Delta S$ -dominated), enabling efficient FAI incorporation and promoting  $\alpha$ -FAPbI<sub>3</sub> phase formation while suppressing mixed-phase generation. This enthalpy–entropy transition underpins the dual role of NBP in enabling humidity-tolerant and scalable perovskite film fabrication.

### Nucleation and crystallization regulation

To comprehensively investigate the effect of NBP on perovskite nucleation and crystallization, we perform in situ time-resolved grazing incidence wide-angle X-ray scattering (GIWAXS) measurements<sup>35–37</sup> (Fig. 2a and Supplementary Fig. 6). The temperature is initially set at 25 °C and heated to 100 °C within 48 seconds. Compared with NMP, NBP produced a stronger signal for the intermediate phase (NBP–PbI<sub>2</sub>, at  $\sim 0.57 \text{ \AA}^{-1}$ ), which rapidly disappeared as the temperature approached 100 °C. By contrast, NMP films exhibited persistent intermediate-phase signals, followed by the emergence of PbI<sub>2</sub> diffraction peaks and low-dimensional polytypes (2H, 4H, 6H), indicative of moisture-induced crystallization defects. Whereas NMP–PbI<sub>2</sub> decayed gradually, NBP–PbI<sub>2</sub> showed a sharp drop in intensity with temperature, reflecting a stronger enthalpic interaction ( $\Delta H$ ) and a more pronounced entropy-driven ( $\Delta S$ ) dissociation.

To further confirm the modulation of perovskite crystallization kinetics by NMP and NBP, time-resolved absorption spectroscopy was conducted (Fig. 2b). The NMP film shows a rapid absorption redshift, while the NBP film shows a notably delayed redshift onset, indicating retarded crystallization kinetics. Kinetic traces at 550 nm (Fig. 2c) reveal that NMP reaches absorption saturation quickly, whereas the NBP film has a slower rise and higher final absorption. These results

confirm that NBP delays perovskite nucleation and crystallization, and yields films with superior crystallinity. The annealed perovskite films finally show that the NBP perovskite film exhibits more ordered crystal orientation (Fig. 2d). We further extract the pole figures (Fig. 2e) from the GIWAXS patterns and calculate the Hermann's factor ( $f$ )<sup>38</sup> (Fig. 2f). The value of  $f = -0.5$  corresponds to the orientation state where the crystal plane is completely parallel to the substrate (that is, exhibiting perfectly anisotropic out-of-plane growth characteristics), while  $f = 0$  indicates an isotropic state of random crystal growth. Thus, when the  $f$  value transitions from  $-0.5$  to  $0$ , it essentially reflects the process by which the crystal growth mode gradually evolves from uniform, ordered out-of-plane orientation to disordered random orientation. Thus, NBP sample ( $f = -0.22253$ ) show more ordered crystalline orientation compared with the NMP sample ( $f = -0.12633$ ).

The XRD analysis (Supplementary Fig. 7 and Supplementary Tables 1 and 2) confirmed that NBP facilitates the preferential growth of high Miller index planes, consistent with the enhanced orientation observed in GIWAXS. This facet regulation reduces defect formation and mitigates moisture-induced degradation. Scanning electron microscopy (SEM) (Supplementary Fig. 8) revealed that NMP films displayed abundant surface voids, which serve as ingress pathways for H<sub>2</sub>O. On exposure to ambient conditions (60% relative humidity, 60 °C) for 24 hours, cross-sectional SEM (Supplementary Fig. 9) showed that NMP films developed additional voids and suffered crystal degradation, whereas NBP films retained their microstructural integrity. Micrometre-scale cross-sectional imaging (Supplementary Fig. 10) further confirmed that NBP facilitates through-thickness grain alignment, which improves charge transport by reducing grain boundary scattering and non-radiative recombination<sup>34</sup>. Moreover, the entropy-driven solvent evaporation promoted by NBP at elevated temperature results in cleaner buried interfaces, enhancing film continuity and device durability. We quantified the residual solvent content in perovskite films using hydrogen nuclear magnetic resonance spectroscopy. The peaks observed at 7.71 ppm and 7.70 ppm and at 2.84 ppm and 2.83 ppm are attributed to the  $-CH$  groups in formamidinium (FA<sup>+</sup>) and the  $-CH_2$  groups in NMP and NBP (Supplementary Fig. 11), respectively. The atomic ratio of NMP to FA<sup>+</sup> was determined to be 8.5%, whereas the corresponding ratio of NBP to FA<sup>+</sup> measured 4.5%. On reaching the 10-minute annealing duration, the residual level of NBP solvent diminishes to 0%, while the remaining concentration of NMP solvent persists at 4.5%, as detailed in Supplementary Tables 3 and 4. Moreover, the fact that the NMP perovskite films exhibit more pores also confirms the above conjecture (Supplementary Figs. 12 and 13).

### Reduction of defect density and sites

To assess the impact of NBP-modified nucleation and crystallization kinetics on scalable, humidity-tolerant perovskite fabrication, we conducted a comprehensive defect-state analysis of the resulting films. As established in previous measurements, NBP promotes enhanced crystallinity while suppressing residual PbI<sub>2</sub> and mixed-phase formation. To further verify this effect, X-ray photoelectron spectroscopy (XPS) was used to probe surface defect chemistry (Fig. 3a). Metallic Pb<sup>0</sup> is known to introduce deep-level trap states within the bandgap, serving as efficient non-radiative recombination centres through the Shockley–Read–Hall mechanism<sup>31,39</sup>, thereby reducing photocarrier lifetimes. Compared with NBP-derived films, NMP samples exhibited notable Pb<sup>0</sup> content, with a Pb<sup>0</sup>/Pb<sup>2+</sup> ratio of 14.8%, confirming the role of NBP in suppressing defect formation during moisture-exposed crystallization. Complementary optical analyses, including photoluminescence quantum yield, steady-state photoluminescence and time-resolved photoluminescence (TRPL), further confirmed the mitigation of non-radiative losses. The photoluminescence quantum yield of the NBP film reached 9.0%, nearly twice that of the NMP sample (4.6%, Supplementary Fig. 14). Consistently, NBP films showed higher photoluminescence intensity (Supplementary Fig. 15) and longer carrier

lifetimes in TRPL measurements (Fig. 3b). To decouple surface and bulk recombination dynamics, we modelled TRPL decay as a function of film thickness using the double heterostructure framework (Fig. 3c and Supplementary Fig. 16):

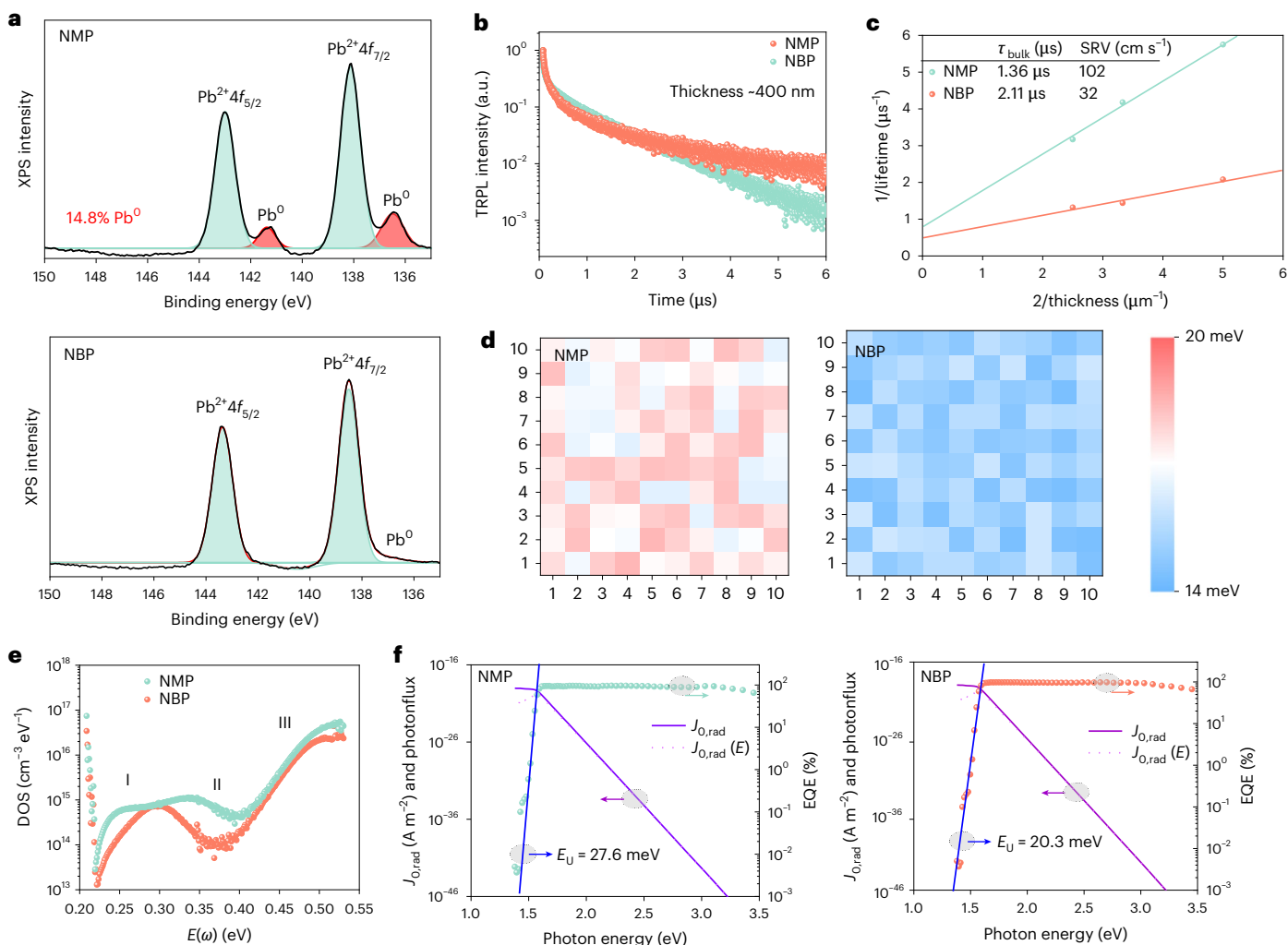
$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_{\text{bulk}}} + \frac{2\text{SRV}}{d}$$

Here  $\tau_{\text{eff}}$  is the effective lifetime of TRPL decay,  $\tau_{\text{bulk}}$  is the bulk lifetime, SRV is the surface recombination velocity and  $d$  is the perovskite thickness<sup>40</sup>. The sample stacking used for the TRPL and SRV measurements is: glass/MeO-2PACz/PIC/perovskite/TOPO, where MeO-2PACz is (2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl)phosphonic acid, PIC is a porous insulator contact composed of Al<sub>2</sub>O<sub>3</sub> islands to reduce interface recombination and TOPO is trioctylphosphine oxide for top-surface passivation<sup>31</sup>. In the NBP sample,  $\tau_{\text{bulk}}$  is substantially reduced to 1.36  $\mu\text{s}$  compared with 2.11  $\mu\text{s}$  in the NMP sample. Concomitantly, the SRV of the NBP film decreases by threefold (from 102 cm s<sup>-1</sup> to 32 cm s<sup>-1</sup>), relative to the NMP sample. These findings indicate that high-humidity ambient conditions induce non-radiative recombination centres at both film surfaces and bulk regions, with degradation initiating at the surface<sup>41,42</sup>. Thus, NBP-mediated enhancement of perovskite crystal coherence and facet order represents a valuable strategy to suppress degradation under high-humidity conditions. Furthermore, photoluminescence and TRPL mapping of a 900- $\mu\text{m}^2$  area for both samples reveals that NBP improves crystal quality, leading to enhanced overall absorption intensity and improved uniformity (Supplementary Figs. 17 and 18). To further evaluate defect levels in NBP perovskite films for large-area printing under high humidity, we quantify Urbach energy ( $E_U$ ), a parameter reflecting Shockley–Read–Hall recombination centre density. The large-area film is uniformly divided into 100 spots via laser scribing (53.29 mm<sup>2</sup> per spot) (Supplementary Fig. 19), and  $E_U$  measurements revealed that NBP suppresses shallow-level defects effectively, with most defects having energies below 17 meV (Fig. 3d). This indicates that improved crystal quality and facet orientation from NBP observably enhance H<sub>2</sub>O resistance by reducing defect-mediated degradation pathways. Compared with the NMP sample, the NBP sample exhibits reduced stress (Supplementary Fig. 20), which contributes to enhanced thermal cycling stability of the device.

Comprehensive defect characterization was further conducted on the devices. As revealed by the trap density of states spectra (Fig. 3e), the integrated trap density of the NBP sample is lower than that of the NMP counterpart. On the basis of the exponential fitting of the external quantum efficiency spectra (Fig. 3f), the NBP sample shows low energy disorder with an  $E_U$  of 20.3 meV. By contrast, an  $E_U$  of 27.6 meV is observed in the NMP sample, which is related to the greater number of defects in the perovskite film. To investigate how NBP modifies trap properties, we use deep-level transient spectroscopy<sup>31</sup>. Trap depths are extracted by linearly fitting deep-level transient spectroscopy peaks to Arrhenius plots (Supplementary Fig. 21). Notably, NBP sample shows only one deep trap (D1) at 0.135 eV above the valence band maximum. By contrast, NMP sample retains D1 and forms two deeper trap states (D2 and D3) at 0.333 eV and 0.538 eV above the valence band maximum. This confirms that NBP reduces deep-level defects by regulating perovskite nucleation, consistent with XPS results. Electrochemical impedance spectroscopy further corroborated these observations (Supplementary Fig. 22).

### Photovoltaic performance and stability

Under AM1.5 G solar illumination, the optimal current density–voltage ( $J$ – $V$ ) characteristics of PSCs and corresponding photovoltaic parameters are shown in Fig. 4a and Supplementary Table 5. Compared with the NMP sample, the NBP sample shows improved performance across all parameters, the PCE is improved from 24.45% to 26.50% and the integrated current density ( $J_{\text{sc}}$ ) from external quantum efficiency also



**Fig. 3 | Reduction of defect density and sites.** **a**, The XPS of Pb 4f for the NMP and NBP perovskite films. The cyan curve denotes the baseline and the orange curve with colour filling represents the ratio of the Pb<sup>0</sup> signal. **b**, The TRPL intensity curve of the NMP and NBP perovskite films with a thickness of ~400 nm. **c**, The SRV and  $\tau_{\text{bulk}}$  are calculated from TRPL decays of perovskite films with varied

thicknesses. **d**, The 100 spots Urbach energy test via laser scribing (53.29 mm<sup>2</sup> per spot) of the NMP and NBP perovskite films. **e**, The total density of states (DOS) spectra. **f**, Exponential fitting of external quantum efficiency (EQE) spectra (Supplementary Fig. 23).

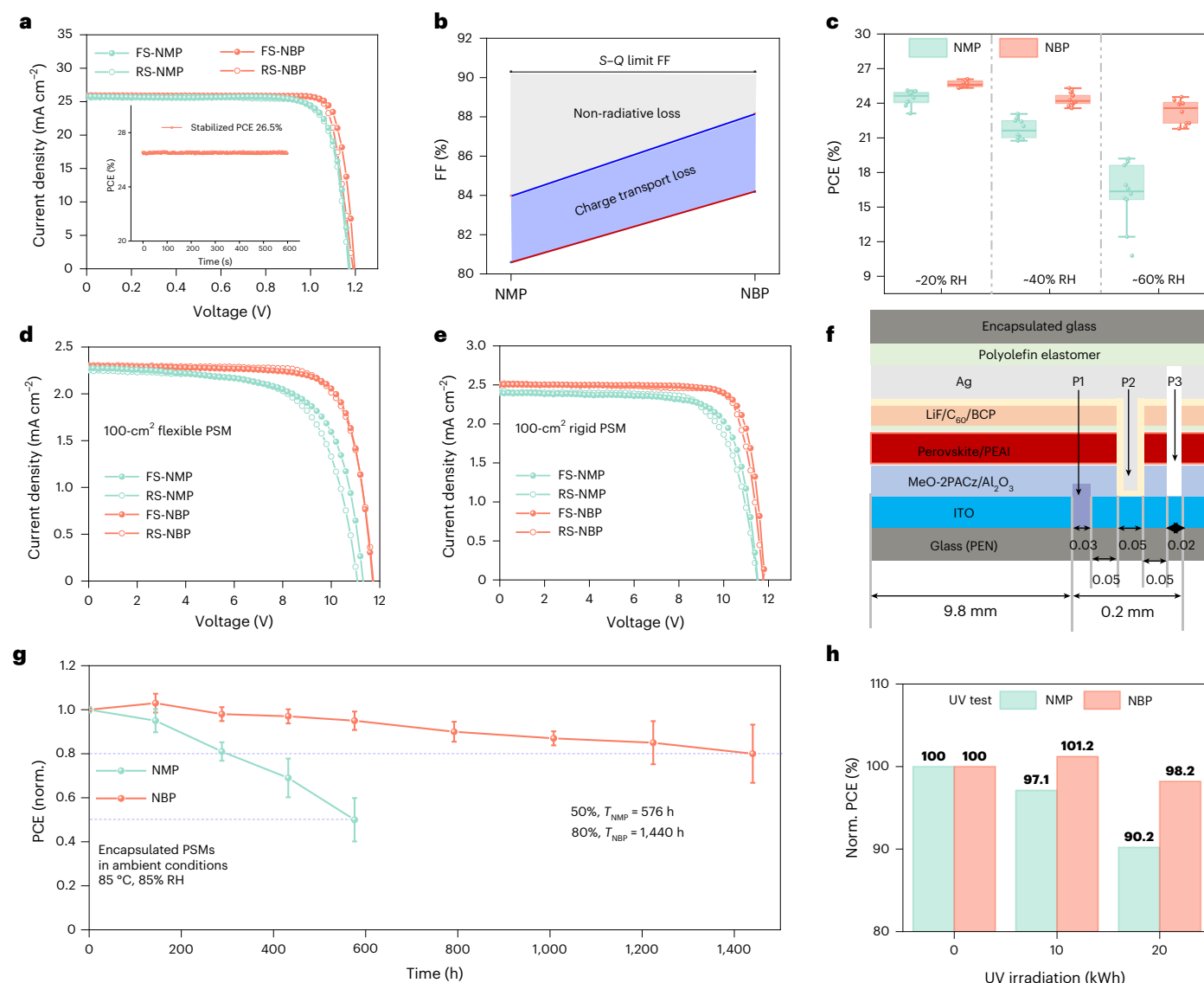
reached ~25 mA cm<sup>-2</sup> (Supplementary Fig. 23). Notably, the fill factor increases from 80.60% to 85.42%. Further light intensity-dependent measurements (Supplementary Fig. 24) reveal that NBP sample shows a slower decrease (1.636 kT/q versus 1.104 kT/q) in open-circuit voltage ( $V_{\text{oc}}$ ) as light intensity decreases. Plotting these data (Fig. 4b) confirms that NBP-promoted ordered perovskite crystal structures observably reduce non-radiative recombination<sup>43</sup>.

To assess batch reproducibility, we fabricated 10 devices per solvent system at relative humidity levels of 20%, 40% and 60% (Fig. 4c). The NBP-based devices displayed high stability and consistency, with the performance gap between NBP and NMP samples widening as humidity increased. When the humidity expanded to 80% relative humidity, 5 NMP samples exhibited no photoelectric performance, while the NBP samples still achieved the highest PCE of up to 23% (Supplementary Fig. 25).

We further demonstrated the scalability of NBP by printing PSMs with an aperture area of 100 cm<sup>2</sup>. The NBP-processed rigid and flexible PSMs achieved PCEs of 24.15% and 20.62%, respectively (Fig. 4d,e and Supplementary Tables 6 and 7), the NMP-processed rigid and flexible PSMs only delivered PCEs of 20.66% and 17.18%. During the fabrication of PSMs with varying aperture areas (Supplementary Fig. 26), we observed that the performance advantage of NBP expands with increasing size,

showing a mere 2.61% loss in PCE at 100 cm<sup>2</sup>, compared with a 4.04% loss in NMP devices. The certification PCE of the rigid and flexible module reaches 23.97% and 19.71% (Supplementary Fig. 27). In addition, we summarize recent reports on ambient printing (Supplementary Table 8) and presented the data in Supplementary Fig. 28. These PCE values correspond to high-performance metrics for scalable perovskite photovoltaic. Also, the subcell efficiencies of NBP modules are more uniform (Supplementary Fig. 29).

To evaluate operational stability, unencapsulated devices were stored under 40% relative humidity. NBP-based devices retained 90.5% of their initial PCE after 2,160 hours, whereas NMP-based devices degraded to 53% within 1,008 hours (Supplementary Fig. 30). And PSM with a configuration of glass or polyethylene naphthalate two formic acid glycol ester (pen)/indium tin oxide (ITO)/MeO-2PACz)/Al<sub>2</sub>O<sub>3</sub>/perovskite/phenethylammonium iodide (PEAI)/lithium fluoride (LiF)/fullerene-C<sub>60</sub>(C<sub>60</sub>)/2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (bathocuproine, BCP)/Ag/polyolefin elastomer/encapsulated glass (Fig. 4f). For encapsulated PSMs (Supplementary Fig. 31), dual 85 testing (85 °C, 85% relative humidity) showed that NBP-based modules retained 80% efficiency after 1,440 hours, while NMP-based counterparts dropped to 50% within 576 hours (Fig. 4g). Moreover, thermal cycling tests conducted between -40 °C and 85 °C showed that the



**Fig. 4 | Photovoltaic performance and stability.** **a**, The PCE of NMP and NBP PSCs. FS, forward scan; RS, reverse scan. **b**, Calculated fill factor (FF) loss of NMP and NBP PSCs. S–Q, Shockley–Queisser. **c**, The PCE for NMP and NBP PSCs under 20%, 40% and 60% relative humidity (RH) at 25 °C (10 PSCs for each condition). Box plots: box limits: 25th–75th percentiles, central line: median,

whiskers: full range (minima to maxima). **d**, The laboratory PCE of flexible PSM. **e**, The laboratory PCE of rigid PSM. **f**, Schematic showing encapsulation and configuration of the PSM. **g**, A double 85 test (85 °C, 85% RH, in ambient air) for thermal stability. Data points and error bars indicate average values and s.d. **h**, UV preconditioning test with a cumulative UV irradiation of 20 kWh m<sup>-2</sup>.

NBP module exhibited no performance degradation after 100 cycles (Supplementary Fig. 32). Under ultraviolet (UV) pretreatment, the NBP module retained 98.2% of its initial performance after 20 kWh m<sup>-2</sup> of total solar irradiation (Fig. 4h), observably exceeding the IEC 61215 standard requirements, whereas the NMP module retained only 90.2% of its initial PCE. These findings collectively demonstrate that solvent engineering using NBP for enthalpy–entropy regulation can enhance perovskite crystal orientation and defect passivation, thereby enabling the fabrication of stable and high-performance devices through scalable ambient-temperature printing processes.

## Conclusion

In summary, we develop an enthalpy–entropy competition strategy for scalable, humidity-tolerant perovskite printing, by using the solvent NBP to suppress the formation of non-ideal phases and PbI<sub>2</sub> under high relative humidity conditions. Compared with NMP, the NBP exhibits a more negative  $\Delta H$  change and a considerable  $\Delta S$  loss. Leveraging its dynamic carbonyl coordination bond, NBP demonstrates stronger

coordination with PbI<sub>2</sub> at room temperature ( $\Delta H$ -dominated process), while it dissociates rapidly at elevated temperatures to promote the incorporation of FAI ( $\Delta S$ -dominated process). This strategy design effectively stabilizes the intermediate phase and reduces the energy barrier within the formamidinium system. Small-area PSCs achieved a PCE of 26.5%, and for 100-cm<sup>2</sup> rigid and flexible PSMs, the certified PCEs were 23.97% and 19.71%. Encapsulated PSMs retain 80% of their initial PCE after 1,440 hours of dual 85 testing (85 °C, 85% relative humidity), demonstrating exceptional humidity-temperature stability. This work offers an approach to mitigate efficiency loss in scalable printed perovskite photovoltaics, leveraging thermodynamics-driven solvent engineering for ambient processing.

## Methods

### Materials

All reagents were used without any further purification. Lead iodide (PbI<sub>2</sub>, 99.999%) was purchased from Alfa Aesar. FAI, lead bromine (PbBr<sub>2</sub>), methylamine bromine, caesium iodide (99.99%), PEAi (99.5%)

and C<sub>60</sub> were purchased from Xi'an Yuri Solar Co., Ltd. MeO-2PACz (purity 98%) was purchased from TCI. Al<sub>2</sub>O<sub>3</sub> dispersions (100 nm, 20 wt% in isopropanol) were purchased from Innocem. BCP and LiF were purchased from Sigma-Aldrich. Anhydrous *N,N*-dimethylformamide (99.8%), NMP (99.7%), NBP (99.7%) and chlorobenzene (99.8%) were purchased from Sigma-Aldrich. Silver (Ag) was purchased from commercial sources with high purity (≥99.999%) and the perovskite composition was Cs<sub>0.01</sub>(FA<sub>0.97</sub>MA<sub>0.03</sub>)<sub>0.99</sub>Pb(I<sub>0.97</sub>Br<sub>0.03</sub>)<sub>3</sub>.

### Preparation of SAMs and Al<sub>2</sub>O<sub>3</sub> layer

PEN/ITO or Glass/ITO substrates (6 Ω sq<sup>-1</sup>) were sequentially cleaned by sonication with detergent, deionized water, acetone and isopropyl alcohol for 20 min, respectively. Then, the self-assembled monolayers (SAMs) were prepared by blading MeO-2PACz (1.5 mg ml<sup>-1</sup>) solution (the solvent was ethanol) at a gap of 50 μm and coating speed of 5 mm s<sup>-1</sup> at room temperature. After printing of SAMs, the substrates had to be annealed at 100 °C for 10 min. Then, the Al<sub>2</sub>O<sub>3</sub> dispersion in isopropanol (1.4 mg ml<sup>-1</sup>) was printed under the same condition and annealed at 100 °C for 10 min to form the passivation layer.

### Device fabrication

All the perovskite film printing was performed by desktop multi-functional and high-precision coating equipment (SSL-BCM-002, Nantong Rollshine Photoelectric Technology Co., Ltd). The perovskite film was fabricated by slot-die coating under ambient conditions. The distance between the slot-die head and the substrate was set at 100 μm, with a relative moving speed of 5 mm s<sup>-1</sup> between the die head and the substrate. The liquid supply rate of the injection pump was controlled at 8 μl s<sup>-1</sup>. The perovskite precursor film was dried with the assistance of vacuum pumping equipment, and the pressure dropped from 10<sup>5</sup> Pa to less than 10 Pa in 10 s. Subsequently, the semi-dried precursor films were annealed first at 70 °C for 1 min and then at 100 °C for 50 min in an ambient form the perovskite film. Then, the PEAI passivation layer was deposited by blade coating 1.5 mg ml<sup>-1</sup> PEAI solution with a speed of 3 mm s<sup>-1</sup>. A complete device was formed by sequentially evaporating LiF (0.1 Å s<sup>-1</sup>, 1 nm), C<sub>60</sub> (0.3 Å s<sup>-1</sup>, 30 nm), BCP (0.1 Å s<sup>-1</sup>, 7 nm) and Ag (0.5 Å s<sup>-1</sup> and 1 Å s<sup>-1</sup>) onto the perovskite film coated with a PEAI passivation layer. For individual PSCs, the thickness of the Ag electrode was 100 nm, whereas for solar cell modules, the Ag layer thickness was 120 nm.

### Fabrication of PSMs

The SAMs/Al<sub>2</sub>O<sub>3</sub>/perovskite/PEAI layers were sequentially deposited on the designated P1 (300 mm s<sup>-1</sup>, 100 kHz) pattern of PEN/ITO as described above. Then, a picosecond laser (365 nm) was used to etch the previously deposited films to form a P2 (300 mm s<sup>-1</sup>, 100 kHz) pattern after LiF and C<sub>60</sub> were sequentially evaporated as described above. Finally, the devices were etched again to form a P3 (250 mm s<sup>-1</sup>, 80 kHz) pattern after evaporation of the BCP and Ag/Cu electrode as described above. The whole module consisted of 10 single cells and the module structure, and the corresponding parameters of P1, P2 and P3 are shown in Fig. 4f, with an active area and geometric fill factor of 98 cm<sup>2</sup> and 98%, respectively.

### Encapsulation of PSMs

We covered the silver electrode surface of the rigid perovskite module with a polyolefin elastomer film to ensure that the polyolefin elastomer film completely covered the module surface, so that the nitrile rubber and the glass substrate could be closely attached. We then placed it in a hot press machine, with a hot-pressing temperature of 120 °C, a pressure of 0.1 MPa and a hot-pressing time of 30 min. After the hot pressing was completed, we let it cool naturally to room temperature. We used sealant to seal the edges of the encapsulated PSM.

### Film characterizations

The morphology of films was measured by atomic force microscopy (nanoscope multimode Bruker) and SEM (JEOL, JSM-7500F, 104) at an

accelerating voltage of 5.0 kV. XRD was recorded by using a D8-Discover 25 diffractometer (Bruker). The steady-state photoluminescence and TRPL measurements at the peak emission of ~810 nm (on the excitation at 553 nm) were carried out by a steady state and lifetime spectrometer (FLS920, Edinburgh Instruments Ltd). The TRPL excitation fluence was ~4 nJ cm<sup>-2</sup> from a 405-nm pulsed laser with a wavelength of 405 ± 8 nm and pulse width of 45 ps, at a repetition rate of 0.1 MHz. The photoluminescence decay data were recorded using time-correlated single photon counting technique. The morphology of the perovskite films was imaged by SEM (JEOL, JSM-7500F). Photoluminescence quantum yield was measured with an Edinburgh FLS1000. The photoluminescence test of large-area perovskite films was conducted by desktop multi-functional and high-precision coating equipment that integrates thickness detection and spectral analysis (SSL-BCM-003, Nantong Rollshine Photoelectric Technology Co., Ltd). We are grateful to the Center of Analytical and Testing Nanchang University for its valuable assistance in XPS testing.

### GIWAXS

GIWAXS data were obtained at beamline BL02U2 of the Shanghai Synchrotron Radiation Facility (SSRF). The monochromatic light source was 1.24 Å. The data were recorded by using the two-dimensional image plate detector of the Pilatus 2M from Dectris, Switzerland. In situ GIWAXS data were collected continuously.

### Device characterizations

The current density–voltage (*J*–*V*) characteristics were generated by the solar simulator (Enlitech, SS-F5-3A, 100 mW cm<sup>-2</sup> irradiance and AM 1.5 G sunlight) in conjunction with the Keithley 2400 Source Meter. The incident photo-to-electron conversion efficiency spectra measurements were detected under monochromatic illumination (Oriel Cornerstone 2601/4m monochromator equipped with an Oriel 70613NS QTH lamp) in a glove box (N<sub>2</sub> atmosphere). The electrical impedance spectroscopy was performed in the dark with a Zahner CIMP5 electrochemical workstation. The frequency was in the range of 0.1 Hz to 10<sup>6</sup> Hz, the amplitude was 50 mV and the d.c. bias was kept at 900 mV. The measurement system (Oriental spectra, NanoQ, NLD640) was equipped with a laser wavelength of 635 nm, a pulse width of <12 ns and a spectral width of 2.7 nm.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

All the data in the paper were derived from our group experiments and are therefore available. These data are published alongside the paper. Source data are provided with this paper.

### References

- Best Research-Cell Efficiency Chart. *National Laboratory of the Rockies* <https://www.nsl.gov/pv/cell-efficiency> (2026).
- Li, S. et al. High-efficiency and thermally stable FACsPbI<sub>3</sub> perovskite photovoltaics. *Nature* **635**, 82–88 (2024).
- Cai, Q. et al. Enhancing electron transport for efficiency-recorded HTL-free inverted perovskite solar cells by molecular complementary passivation. *Joule* **9**, 101880 (2025).
- Luo, C. et al. Engineering bonding sites enables uniform and robust self-assembled monolayer for stable perovskite solar cells. *Nat. Mater.* **24**, 1265–1272 (2025).
- Liu, D. et al. Strain relaxation in halide perovskites via 2D/3D perovskite heterojunction formation. *Sci. Adv.* **11**, eadu3459 (2025).
- Li, J. et al. Homogeneous coverage of the low-dimensional perovskite passivation layer for formamidinium-caesium perovskite solar modules. *Nat. Energy* **9**, 1540–1550 (2024).

7. Chu, Z. et al. Synergistic macroscopic-microscopic regulation: dual constraints of the island effect and coffee-ring effect in printing efficient flexible perovskite photovoltaics. *Adv. Funct. Mater.* **35**, 2424191 (2025).
8. Fan, B. et al. A bionic interface to suppress the coffee-ring effect for reliable and flexible perovskite modules with a near-90% yield rate. *Adv. Mater.* **34**, 2201840 (2022).
9. Zhao, X. et al. Operationally stable perovskite solar modules enabled by vapor-phase fluoride treatment. *Science* **385**, 433–438 (2024).
10. Li, D. et al. Selection of functional spacer cations for efficient 2D/3D perovskite solar cells. *CCS Chem.* **5**, 781–801 (2023).
11. Xu, Y. et al. Printable oil-NiO<sub>x</sub> hole transport layer enables efficient n-i-p perovskite solar cells with a high thermal stability. *Sci. China Chem.* **67**, 2335–2340 (2024).
12. Wang, H. et al. Impurity-healing interface engineering for efficient perovskite submodules. *Nature* **634**, 1091–1095 (2024).
13. Xing, Z. et al. Multi-environment phase stabilization by lattice reinforcement for efficient perovskite solar cells. *Sci. China Mater.* **66**, 2573–2581 (2023).
14. Yan, K. et al. Hybrid halide perovskite solar cell precursors: colloidal chemistry and coordination engineering behind device processing for high efficiency. *J. Am. Chem. Soc.* **137**, 4460–4468 (2015).
15. Yaghoobi Nia, N. et al. Solution-based heteroepitaxial growth of stable mixed cation/anion hybrid perovskite thin film under ambient condition via a scalable crystal engineering approach. *Nano Energy* **69**, 104441 (2020).
16. Bu, T. et al. Lead halide-templated crystallization of methylamine-free perovskite for efficient photovoltaic modules. *Science* **372**, 1327–1332 (2021).
17. Zou, Y. et al. A crystal capping layer for formation of black-phase FAPbI<sub>3</sub> perovskite in humid air. *Science* **385**, 161–167 (2024).
18. Wang, L. et al. A Eu<sup>3+</sup>-Eu<sup>2+</sup> ion redox shuttle imparts operational durability to Pb-I perovskite solar cells. *Science* **363**, 265–270 (2019).
19. Steele, J. A. et al. Direct laser writing of  $\delta$ - to  $\alpha$ -phase transformation in formamidinium lead iodide. *ACS Nano* **11**, 8072–8083 (2017).
20. Hidalgo, J. et al. Synergistic role of water and oxygen leads to degradation in formamidinium-based halide perovskites. *J. Am. Chem. Soc.* **145**, 24549–24557 (2023).
21. Kim, H. et al. Optimal interfacial engineering with different length of alkylammonium halide for efficient and stable perovskite solar cells. *Adv. Energy Mater.* **9**, 1902740 (2019).
22. Chen, L. et al. Intrinsic phase stability and inherent bandgap of formamidinium lead triiodide perovskite single crystals. *Angew. Chem. Int. Ed.* **134**, e202212700 (2022).
23. Feng, W. et al. Near-stoichiometric and homogenized perovskite films for solar cells with minimized performance variation. *Angew. Chem. Int. Ed.* **135**, e202300265 (2023).
24. Yan, L. et al. Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt. *Nat. Energy* **8**, 1158–1167 (2023).
25. Wan, Z. et al. Efficient perovskite solar modules with an ultra-long processing window enabled by cooling stabilized intermediate phases. *Energy Environ. Sci.* **17**, 6302–6313 (2024).
26. Liu, P. et al. Ambient scalable fabrication of high-performance flexible perovskite solar cells. *Energy Environ. Sci.* **17**, 7069–7080 (2024).
27. Jeon, N. J. et al. Solvent engineering for high-performance inorganic-organic hybrid perovskite solar cells. *Nat. Mater.* **13**, 897–903 (2014).
28. Chao, L. et al. Solvent engineering of the precursor solution toward large-area production of perovskite solar cells. *Adv. Mater.* **33**, 2005410 (2021).
29. Duan, C. et al. Scalable fabrication of wide-bandgap perovskites using green solvents for tandem solar cells. *Nat. Energy* **10**, 318–328 (2024).
30. Wang, Y. et al. Solvent-assisted reaction for spontaneous defect passivation in perovskite solar cells. *Nat. Photonics* **19**, 985–991 (2025).
31. Meng, H. et al. Inhibition of halide oxidation and deprotonation of organic cations with dimethylammonium formate for air-processed p-i-n perovskite solar cells. *Nat. Energy* **9**, 536–547 (2024).
32. Qin, M. et al. Manipulating the mixed-perovskite crystallization pathway unveiled by in situ GIWAXS. *Adv. Mater.* **31**, 1901284 (2019).
33. Ma, C. et al. Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells. *Science* **379**, 173–178 (2023).
34. Li, S. et al. Coherent growth of high-miller-index facets enhances perovskite solar cells. *Nature* **635**, 874–881 (2024).
35. Li, H. et al. Optimizing the crystallization behavior and film morphology of donor-acceptor conjugated semiconducting polymers by side-chain-solvent interaction in nonpolar solvents. *Macromolecules* **54**, 10557–10573 (2021).
36. Gao, W. et al. Efficient all-small-molecule organic solar cells processed with non-halogen solvent. *Nat. Commun.* **15**, 1946 (2024).
37. Ma, R. et al. In-situ understanding on the formation of fibrillar morphology in green solvent processed all-polymer solar cells. *Natl Sci. Rev.* **11**, nwae384 (2024).
38. Liu, H. et al. Simultaneous mechanical and chemical synthesis of long-range-ordered perovskites. *Nat. Synth.* **4**, 196–208 (2025).
39. Han, C. et al. Program-modulated kinetics of perovskite-film growth by molecular ‘thruster’ for high-efficiency and stable perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202419726 (2025).
40. Tong, J. et al. Carrier control in Sn-Pb perovskites via 2D cation engineering for all-perovskite tandem solar cells with improved efficiency and stability. *Nat. Energy* **7**, 642–651 (2022).
41. Stolterfoht, M. et al. Visualization and suppression of interfacial recombination for high-efficiency large-area pin perovskite solar cells. *Nat. Energy* **3**, 847–854 (2018).
42. Li, B. et al. Fundamental understanding of stability for halide perovskite photovoltaics: the importance of interfaces. *Chem* **10**, 35–47 (2024).
43. Zhan, L. et al. Anchorable polymers enabling ultra-thin and robust hole-transporting layers for high-efficiency inverted perovskite solar cells. *Angew. Chem. Int. Ed.* **137**, e202422571 (2025).

## Acknowledgements

A portion of this work is based on the data obtained at Beijing Synchrotron Radiation Facility (BSRF) and SSRF. We thank the 1W1A-Diffuse X-ray Scattering Beamline of BSRF (<https://cstr.cn/31109.02.BSRF.1W1A>) and BL16B1, BL02U2 and BL03HB of SSRF for providing technical support and assistance in GIWAXS data collection.

## Author contributions

X.H. and Y.C. directed and supervised the project. Z.C. and X.H. conceived of and designed the experiments. Z.C. and X.H. completed the writing of the paper. Z.C., J.L., Y.X., Z.G. and J.G. fabricated the PSMs. Z.C., S.L., B.F. and Y.L. fabricated the PSCs. Z.C., J.L., J.G. and H.L. characterized the GIWAXS. Z.C., Z.X., Z.G. and X.M. characterized the various photoelectric properties. All authors contributed to discussions and finalizing the paper.

## Funding

We thank the following for support: National Key Research and Development Program of China (grant nos. 2024YFF1401100 and 2024YFB3815200), the National Natural Science Foundation of China

(NSFC) (grant nos. 52527804, 52573277, 52403323 and 22461142139), the Natural Science Foundation of Jiangxi Province (grant no. 20242BAB24002), Hebei Province Central Guidance Fund for Local Science and Technology Development (grant no. 254Z4301G), Nanchang University Interdisciplinary Research Funding Program (grant no. 202505300006), Shenzhen Science and Technology Program (grant no. JCYJ20241202124937050) and the State Key Laboratory for Mechanical Behavior of Materials.

## Competing interests

The authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s44160-026-01142-0>.

**Correspondence and requests for materials** should be addressed to Xiaotian Hu or Yiwang Chen.

**Peer review information** *Nature Synthesis* thanks Yonghua Chen, Jingrui Li and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Alexandra Groves, in collaboration with the *Nature Synthesis* team.

**Reprints and permissions information** is available at [www.nature.com/reprints](http://www.nature.com/reprints).

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

## Solar Cells Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our [data availability policy](#), see [Authors & Referees](#).

### ► Experimental design

#### Please check: are the following details reported in the manuscript?

##### 1. Dimensions

- Area of the tested solar cells ☒ Yes Described in main text 0.1 cm<sup>2</sup>, 100 cm<sup>2</sup>  
☐ No
- Method used to determine the device area ☒ Yes the device area are defined by aperture  
☐ No

##### 2. Current-voltage characterization

- Current density-voltage (J-V) plots in both forward and backward direction ☐ Yes The devices have been tested in forward and reverse direction. Data reported in Fig. 4a.  
☒ No
- Voltage scan conditions ☐ Yes For the forward scan From negative to positive and for the reverse scan from positive to negative with the scan step length of 0.02 V.  
*For instance: scan direction, speed, dwell times* ☒ No
- Test environment ☐ Yes After the devices are prepared, they are tested in a glove box (nitrogen atmosphere, 25 °C).  
*For instance: characterization temperature, in air or in glove box* ☒ No
- Protocol for preconditioning of the device before its characterization ☐ Yes The devices were not pretreated prior to characterization.  
☒ No
- Stability of the J-V characteristic ☒ Yes Provide in Supplementary Fig. 4a.  
*Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see [ref. 7](#) for details.* ☐ No

##### 3. Hysteresis or any other unusual behaviour

- Description of the unusual behaviour observed during the characterization ☐ Yes No unusual behaviour.  
☒ No
- Related experimental data ☐ Yes No unusual behaviour.  
☒ No

##### 4. Efficiency

- External quantum efficiency (EQE) or incident photons to current efficiency (IPCE) ☒ Yes Provide in Supplementary Fig. 23.  
☐ No
- A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator ☒ Yes Provide in Supplementary Fig. 23.  
☐ No
- For tandem solar cells, the bias illumination and bias voltage used for each subcell ☐ Yes No tandem solar cells are involved in this manuscript.  
☒ No

##### 5. Calibration

- Light source and reference cell or sensor used for the characterization ☒ Yes Described in experimental section.  
☐ No
- Confirmation that the reference cell was calibrated and certified ☒ Yes Described in experimental section.  
☐ No

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes  
☐ No

Described in experimental section.

## 6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes  
☐ No

100 cm<sup>2</sup>

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes  
☒ No

This part has been studied in our previous work

## 7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes  
☐ No

South China National Center of Metrology.

A copy of any certificate(s)

*Provide in Supplementary Information*

☒ Yes  
☐ No

Provide in Supplementary Fig. 27.

## 8. Statistics

Number of solar cells tested

☒ Yes  
☐ No

Provide in Fig. 4c and Supplementary Fig. 25.

Statistical analysis of the device performance

☒ Yes  
☐ No

Provide in Fig. 4c and Supplementary Fig. 25.

## 9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

*For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature*

☒ Yes  
☐ No

Provide in Fig. 4g,h and Supplementary Fig. 30, 32